Python Tuple Datatype

A single or group of elements which are enclosed between parenthesis or without parenthesis and separated by comma is called as tuple in python

--> For Example t = (10, 20, 30) t2 = 10, 20, 30

Note: Inside tuple mutable and immutable objects are allowed

-> Tuples can also be created with a single element, but it is a tricky

-) Having one element in the parenthesis is not sufficient, there must be a 'comma' to make it a tuple

+ we can create single element tuple with comma separation

t = 10, t2 = (10,)

-) we can create multiple element tuple with comma separation

t1 = (10, 20, 30) & t2 = 10, 20, 30

--> Homogenous tuple object (10, 20, 30, 40, 50)

-) Heterogenous tuple object --> (10, 20.50, True, 'Hi', [1, 2, 3], {1, 2})

-) tuple datatype represents using tuple type ex: t = (1,2,3,4)

print (type(t))

* * *

we can create empty tuple object by using empty parentheses or tuple () function

for example t1 = () t2 = tuple ()

7) Tuple is Immutable type object

Important Points

* tuple object follows insertion order
* tuple object allows duplicate values
* every element in the tuple object is represented with unique index
* python support both forward and backward indexes
* forward index starts from 0
* Backward index starts from -1

for example

```
        0    1    2   3   4   5   6   7   8
s = (10, 20, 30, 40, 50, 60, 70, 80, 'hai')
     -9   -8   -7  -6  -5  -4  -3  -2  -1
```

* slicing --> object [start-index-value : end-index-value : step-value]

Here

* by default start index value is 0,

* by default end index value is object lenght

* by default step value is 1

* Slicing ---> object [start_index_value : end_index value : step_valu

* If step value is negative end index value 'n+1'
" " " postive " " " n-1

* If step value is positive is there then start index value start index value will be low and end index value will be high

* Python tuple supports " concatenation " behaviour

(10, 20) + (30,40)

10, 20, 30, 40

* Python tuple supports " repetion behaviour

(10, 20) * 3

out (10, 20, 10, 20, 10, 20)

* Python Tuple supports tuple unpacking (extracting given object elements automatically)

a, b, c = (10, 20, 30)

a = 10 b = 20 c = 30